

1 Features list

Categorized Features

This section shows the final list of features that have been used for predicting burned area and fire occurrence.

```
id_encoder
rhum_mean
Past_risk
fwi_max
cluster_encoder
isi_mean
rhum_max
bui_max
ffmc_mean
ffmc_max
dayofyear
sum_rain_last_7_days_mean
nesterov_mean
NDSI_min
calendar_sum
calendar_min
prec24h_max
angstroem_mean
sum_rain_last_7_days_max
munger_mean
dc_max
month
calendar_max
nesterov_max
rhum16_mean
precipitationIndexN5_max
precipitationIndexN5_mean
munger_max
angstroem_max
Pin_maritime_max
precipitationIndexN3_max
foret_encoder_mean
Past_burnedarea
rhum_min
days_since_rain_max
temp_max
rhum16_max
Road_mean
```

temp16_max
fwi_min
NDSI_mean
sum_rain_last_7_days_min
prec24h16_max
dailySeverityRating_min
ffmc_min
bui_min
days_since_rain_min
Corine_vegetation_mean
rhum16_min
foret_encoder_max
Corine_vegetation_max
Chênes sempervirents_max
holidays
prcp_max
wdir16_mean
precipitationIndexN5_min
NDMI_min
munger_min
nesterov_min
Corine_grass_mean
wspd16_min
angstroem_min
Corine_urban_max
Mélèze_max
wdir_max
temp16_min
dc_min
dwpt_max
wdir_mean
population_mean
Pins mélangés_mean
Corine_littoral_max
Peuplier_max
dwpt16_max
NC_mean
NDMI_max
Mélèze_mean
Châtaignier_max
Pin laricio, pin noir_max
dwpt_min
NDSI_max
dwpt16_min
Corine_agricultural_mean
NDWI_max

Pins mélangés_max
wdir_min
NDMI_mean
Corine_water_mean
Pin d'Alep_max
Corine_grass_max
elevation_max
Corine_transport_mean
Conifères_mean
elevation_mean
wspd_max
NC_max
Pin autre_max
Peuplier_mean
prcp16_max
Douglas_mean
corine_encoder_mean
precipitationIndexN3_min
Feuillus_max
NR_max
Corine_rock_max
holidaysBorder
population_max
NDVI_mean
Pin à crochets, pin cembro_max
Sapin, épicéa_max
Conifères_max
NDVI_max
Pin sylvestre_mean
Mixte_max
Mixte_mean
Robinier_max
Pin laricio, pin noir_mean
wspd_mean
Douglas_max
wdir16_min
Corine_forest_mean
Pin sylvestre_max
Robinier_mean
wdir16_max
Chênes décidus_max
Corine_transport_max
Corine_water_max
Chênes décidus_mean
elevation_min
kbdi_max

wspd16_mean
Feuillus_mean
Corine_forest_max
Hêtre_mean
wspd_min
Hêtre_max
wspd16_max
prec24h16_min
prec24h_min
Corine_agricultural_max
corine_encoder_min
foret_encoder_min
ramadan
Corine_moisture_max
dayofweek
snow24h_max
snow24h16_max
sum_snow_last_7_days_max
isweekend
population_min
sum_snow_last_7_days_min
sum_consecutive_rainfall_max
prcp16_min
prcp_min
Road_min
snow24h_min
confinement
Feuillus_min
kbdi_min
sum_consecutive_rainfall_min
snow24h16_min
couvrefeux

2 Tree based configuration

Table 1 presents the configuration of tree based models.

3 Deep learning layers configuration

This section present the layers parameters for each deep learning model used in this study.

3.1 MLP

Table 2 shows the configuration of the MLP models.

Table 1: **Configuration of each tree based model**

Parameters	XGBoost	Catboost
Early Stopping Rounds	15	15
Learning Rate	0.001	0.001
Max Depth/Num Leaves	6	4
Min Child Weight	1.0	-
Max Delta Step	1.0	-
Subsample	0.5	0.7
Col Sample Bytree	0.7	0.6
Reg Lambda	1.7	1
Reg Alpha	0.7	0.27
N Estimators	10000	10000
Tree Method	hist	-

Table 2: Layer-wise architecture of the MLP model

Layer	Configuration / Dimensions
Linear 1	Linear(162 $\rightarrow$ 162), ReLU
Linear 2	Linear(162 $\rightarrow$ 162), ReLU
Linear 3	Linear(162 $\rightarrow$ 64), ReLU
Linear 4	Linear(64 $\rightarrow$ 5)

3.2 GraphCastGRY

Table 3 shows the configuration of the GraphCast model. Edges channels is set to 4 as in the original paper containing the x, y, z positions of the mesh node and the length of the edge. Similarly, the input mesh data is the 3D position of the node.

Table 3: Layer-wise architecture of the **GraphCastGRU** model

Layer	Configuration / Dimensions
GRU	GRU(num_layers=2, input_size= <code>in_channels</code> , hidden_size=128)
BatchNorm	BatchNorm1d(128)
Dropout	Dropout(0.03)
GraphCastNet	Message Passing (6 layers), hidden dim = 128
Linear 1	Linear(128 $\rightarrow$ 256)
Linear 2	Linear(256 $\rightarrow$ 64)
Output Layer	Linear(64 $\rightarrow$ <code>out_channels</code>)
Activation (internal)	ReLU
Activation (output)	Softmax (for classification)

3.3 GRU

Table 4 shows the configuration of the GRU model.

Table 4: Layer-wise architecture of the GRU model

Layer	Configuration / Dimensions
GRU	2 layers, hidden size 162, ReLU, dropout 0.03
Norm 1	BatchNorm
Dropout	0.03
Linear 1	Linear(162 $\rightarrow$ 256), ReLU
Linear 2	Linear(256 $\rightarrow$ 64), ReLU
Linear 3	Linear(64 $\rightarrow$ 5)

3.4 LSTM

Table 5 shows the configuration of the LSTM model.

Table 5: Layer-wise architecture of the LSTM model

Layer	Configuration / Dimensions
LSTM	1 layer, hidden size 64, ReLU
Norm 1	BatchNorm
Dropout	0.03
Linear 1	Linear(64 $\rightarrow$ 64), ReLU
Linear 2	Linear(64 $\rightarrow$ 64), ReLU
Linear 3	Linear(64 $\rightarrow$ 5)

3.5 DilatedCNN

Table 6 shows the configuration of the DilatedCNN model.

Table 6: Layer-wise architecture of the DilatedCNN model (Dil = dilation)

Layer	Configuration / Dimensions
Conv 1	Conv1d(162 filters, dilation 1)
Conv 2	Conv1d(162 filters, dilation 3)
Conv 3	Conv1d(dilation 3)
Norm + Dropout	BatchNorm, dropout 0.03
Linear 1	Linear(162 $\rightarrow$ 64)
Linear 2	Linear(64 $\rightarrow$ 64)
Linear 3	Linear(64 $\rightarrow$ 5)

4 Features importance

This section features importance based on shap values for Catboost models. Figure 1 shows features importance of Burned area prediction of Catboost model.

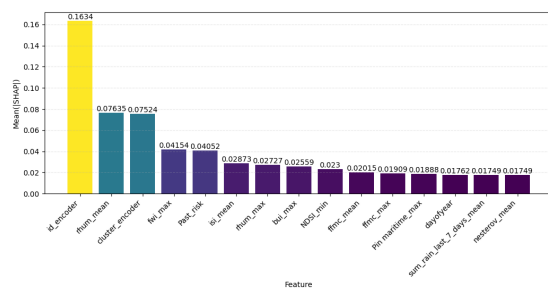


Figure 1: Top 15 shap values computed on multi-classification Catboost model for Burned area prediction. ID encoder correspond to Department ID